

Mid Semester Examination 2025-26

Name of the Program: B.Tech. Biotechnology
Name of the Course: Fermentation Technology

Semester: V
Course Code: TBT 503

Time: 90 min

Maximum Marks: 50

SET-2

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Define fermentation technology and discuss the historical development of the fermentation industry.	
OR		
(b)	Explain with examples how microbial culture selection influences product yield. Analyze the role of various parameters.	
Q2	(10 marks)	CO1
(a)	How do you isolate any strain from soil and identify it? Highlight methods to preserve it.	
OR		
(b)	Discuss various stages in scale up of fermentative process from lab scale to downstream processing stage.	
Q3	(10 marks)	CO2
(a)	Apply mass balance and Energy Balance Equation on a process running in continuous culture mode and assume that it is exothermic process and work is done on the system with loss of heat.	
OR		
(b)	If air consists of 77% by weight of nitrogen and 23% by weight of oxygen calculate: (a) the mean molecular weight of air, (b) the mole fraction of oxygen, (c) the concentration of oxygen in mole/m ³ and kg/m ³ if the total pressure is 1.5 atmospheres and Temperature is 25 °C. (Assume 100 kg of Air).	
Q4	(10 marks)	CO2
(a)	Aerobic growth of <i>S. cerevisiae</i> on ethanol is simply described by the following overall reaction: $C_2H_5OH + a O_2 + b NH_3 \rightarrow c CH_{1.7}O_{4.9}N_{0.149} + d CO_2 + e H_2O$ a. Determine the coefficients a, b, c, and d, where RQ = 0.66. b. Determine the biomass yield coefficient Y_{x/O_2} and oxygen yield coefficient Y_{O_2/O_2} (gdw/g O_2).	
OR		
(b)	Aerobic degradation of an organic compound by a mixed culture of organisms in waste water can be represented by the following reaction: $C_3H_6O_3 + a O_2 + b NH_3 \rightarrow c C_5H_7NO_2 + d H_2O + e CO_2$	

	a. Determine a, b, c, d, and e, if $Y_x/\square = 0.4 \text{ g X/g S}$. b. Determine the yield coefficients $Y_x/O\square$ and $Y_x/NH\square$.	
Q5	(10 marks)	CO2
(a)	What do you mean by aseptic transfer? Enlist various factors that affect the inoculum development of the fungus.	
OR		
(b)	Derive an expression for the 3 types of products obtained in Batch culture. Plot the graph for each case.	